

BF Service on NIST AWS

Bugs Framework (BF)
<https://usnistgov.github.io/BF>

Cloud Computing Day
08/13/2026

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BF Service

- BF Tool App
 - ✓ Create, save, edit, graph, upload, download, parse formal BF Vulnerability Specifications
- **30+** UIs & APIs
 - ✓ BF Functionalities – parse, graph, backward trace
 - ✓ BF datasets – generate & filter BFCWEs, BFCVEs, Security Rules, Descriptions, BF Taxonomies, LL(1) CFG, Relations, Transitions, Operations Flow

Infrastructure & Technology Stack

- Cloud Platform: NIST AWS Environment
- Operating System: Windows Server 2025
- Compute Environment: Amazon EC2
- Application Layer: ASP.NET Core MVC / .NET 10 (C#)
- Database Layer: SQLite or MS SQL Server

Access Control

- Authentication
 - ✓ NIST Okta (for NIST researchers)
 - ✓ API Keys support ready
- Authorization
 - ✓ BF-specific Roles
 - ✓ Across APPs, UIs & APIs

Potential Impact

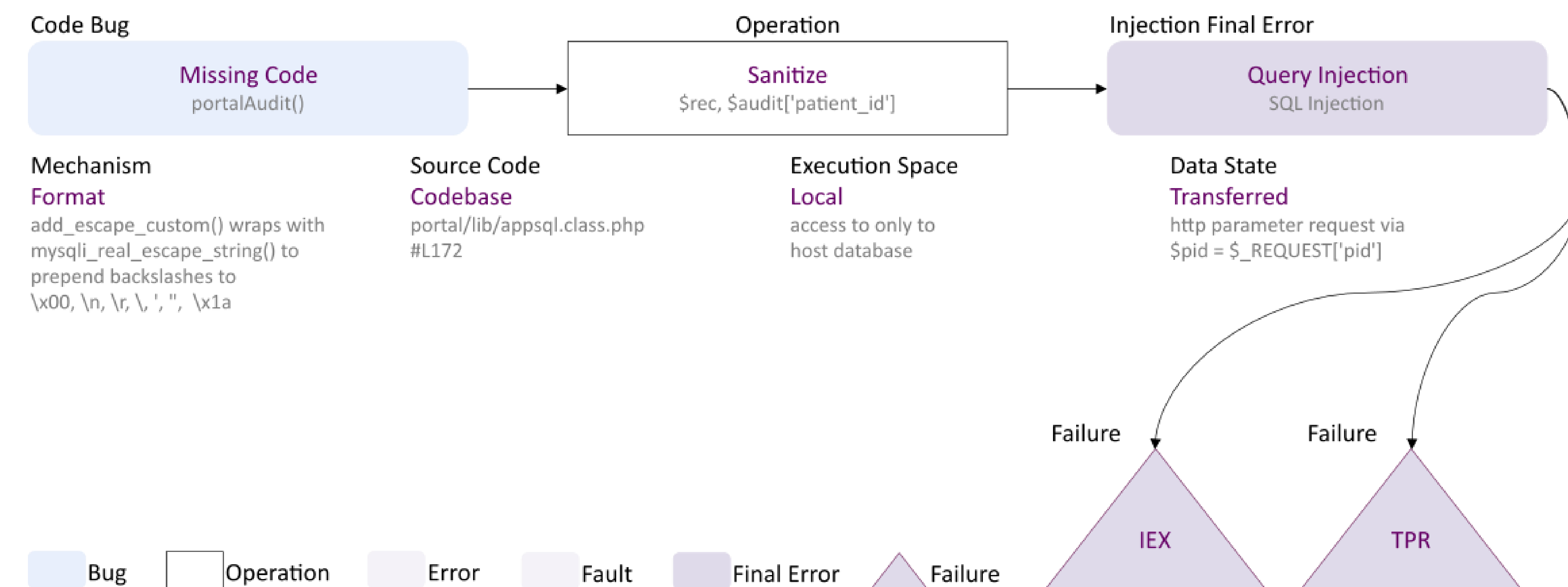
- Direct access to centralized BF functionalities
- Real-time access to shared BF datasets
- Collaborative NVD^{BF} augmentation
- Highly informed cybersecurity R&D innovations

Example
BF Vulnerability Specification via BF Tool APP
& Generation of Graphical Representation

BF Specification of CVE-2018-17181
SQL Injection in OpenEMR before 5.0.1 Patch 7

Input/Output Check (_INP)
Data Validation (DVL)

Injection Final Error



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